

EE 671: VLSI Design

ASSIGNMENT 1



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Objective

The objective of this experiment is to design and characterize a minimum size CMOS inverter (INVX1) with equal rise and fall times and a channel length of 0.15 μm . A strength-2 inverter (INVX2) is also designed and analyzed under similar conditions.

Design Parameters and Dynamic Characteristics

Minimum Size Inverter (INVX1)

Table 1: Design Parameters of INVX1

Parameter	Value
PMOS Width (μm)	1.26
PMOS Length (μm)	0.15
NMOS Width (μm)	0.42
NMOS Length (μm)	0.15

Table 2: Transient Characteristics of INVX1

Characteristic	Value
Rise time, t_r (ps)	18.524
Fall time, t_f (ps)	18.519
Propagation delay, t_p (ps)	21.280

Table 3: DC Characteristics of INVX1

Parameter	Value
V_{IH} (V)	1.006
V_{IL} (V)	0.774
NM_H (V)	0.794
NM_L (V)	0.774
Switching Voltage, V_M (V)	0.893

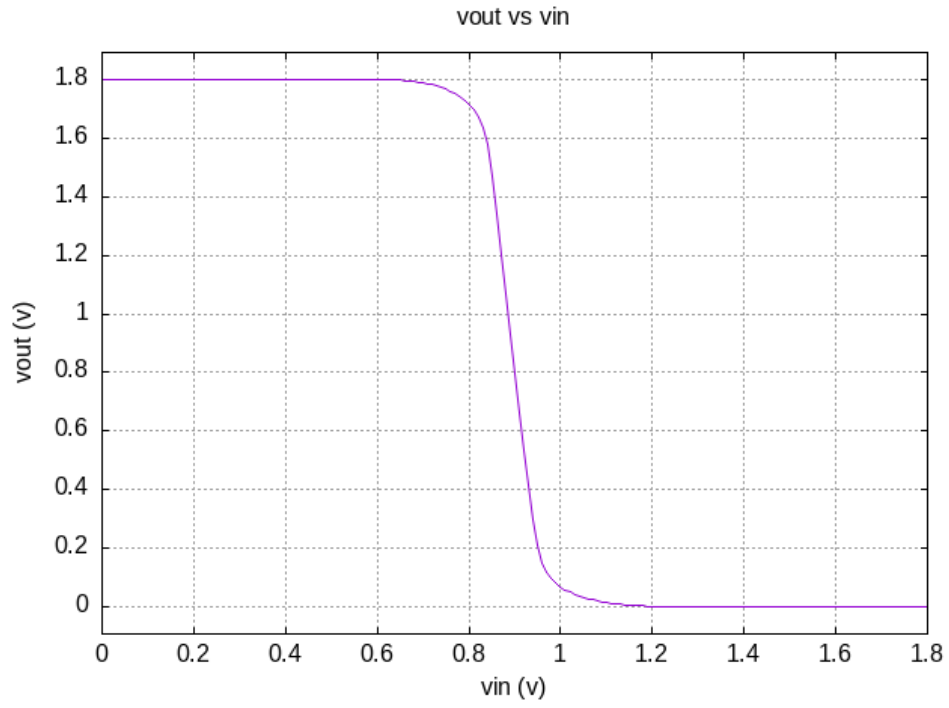


Figure 1: DC transfer curve of INVX1.

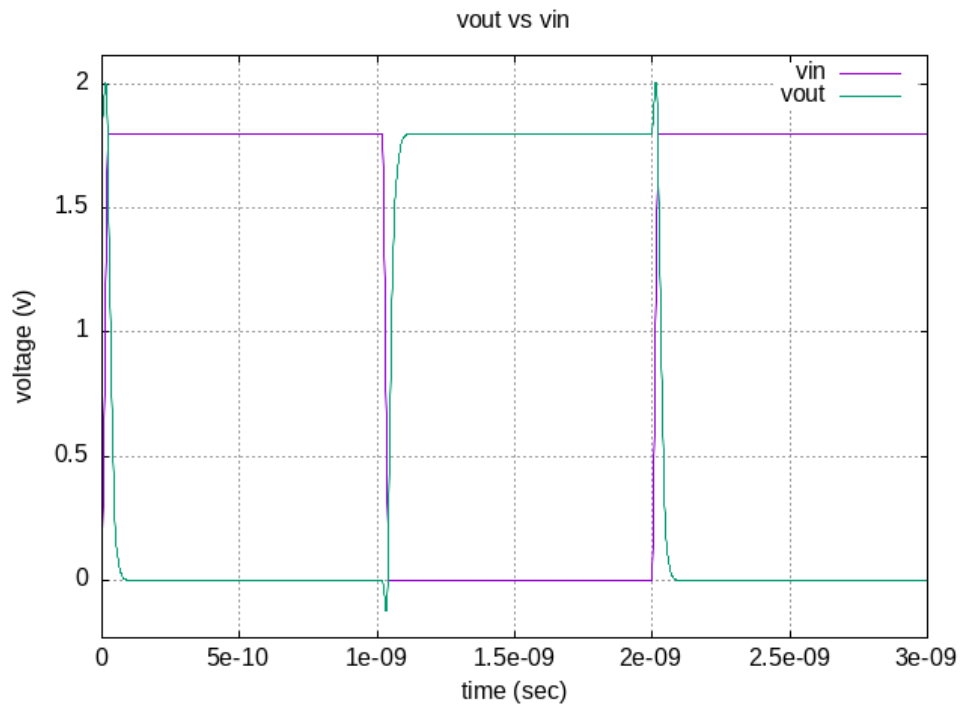


Figure 2: V_{out} vs V_{in} for INVX1.

Strength-2 Inverter (INVX2)

Table 4: Design Parameters of INVX2

Parameter	Value
PMOS Width (μm)	2.57
PMOS Length (μm)	0.15
NMOS Width (μm)	0.84
NMOS Length (μm)	0.15

Table 5: Transient Characteristics of INVX2 (loaded with INVX1)

Characteristic	Value
Rise time, t_r (ps)	11.614
Fall time, t_f (ps)	11.611
Propagation delay, t_p (ps)	16.376

Table 6: DC Characteristics of INVX1

Parameter	Value
V_{IH} (V)	1.020
V_{IL} (V)	0.767
NM_H (V)	0.780
NM_L (V)	0.767
Switching Voltage, V_M (V)	0.895

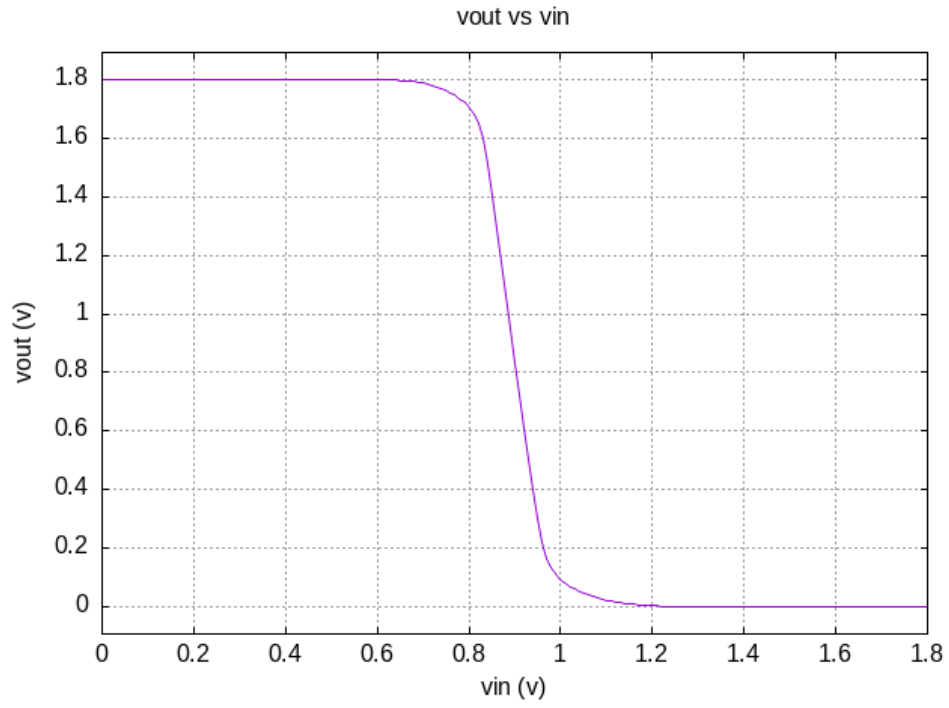


Figure 3: DC transfer curve of INVX2.

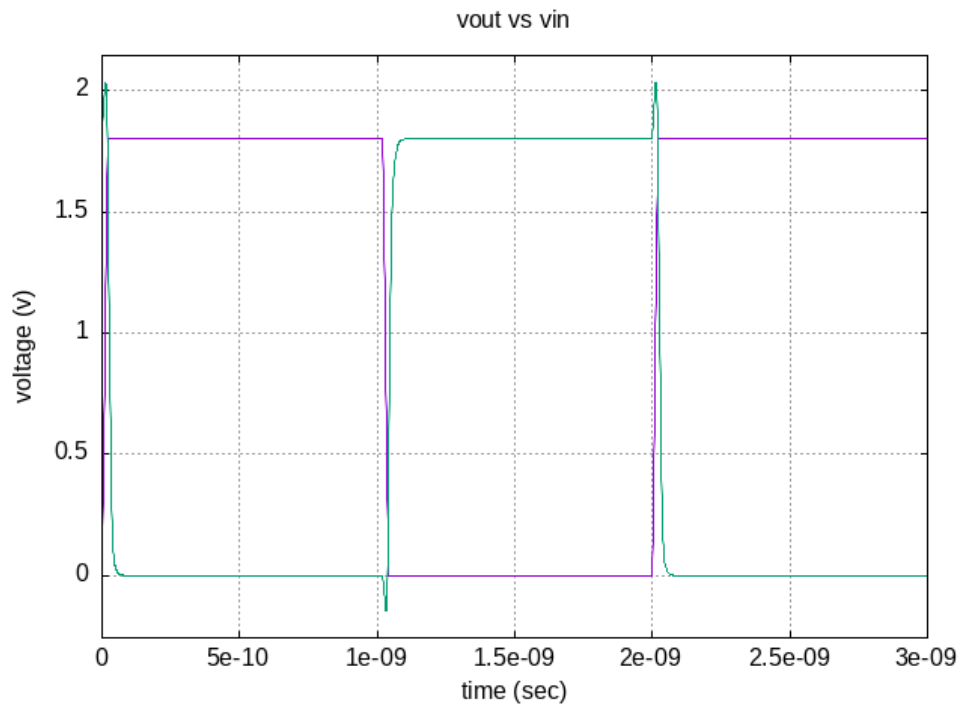


Figure 4: V_{out} vs V_{in} for INVX2.

Code

Listing 1: NGSpice netlist for INVX1 (not1.cir)

```
.subckt not1 vin vout vdd vss l=0.15 wp=1.26 wn=0.42
xmp vout vin vdd vdd sky130_fd_pr__pfet_01v8 l={l} w={wp} as={2*{wp}
    }*{l}} ad={2*{wp}*{l}} ps={2*({wp} + 2*{l})} pd={2*({wp} + 2*{l})
    }
xmn vout vin vss vss sky130_fd_pr__nfet_01v8 l={l} w={wn} as={2*{wn}
    }*{l}} ad={2*{wn}*{l}} ps={2*({wn} + 2*{l})} pd={2*({wn} + 2*{l})
    }
.ends
```

Listing 2: NGSpice netlist for INVX2 (not2.cir)

```
.subckt not2 vin vout vdd vss l=0.15 wp=2.57 wn=0.84
xmp vout vin vdd vdd sky130_fd_pr__pfet_01v8 l={l} w={wp} as={2*{wp}
    }*{l}} ad={2*{wp}*{l}} ps={2*({wp} + 2*{l})} pd={2*({wp} + 2*{l})
    }
xmn vout vin vss vss sky130_fd_pr__nfet_01v8 l={l} w={wn} as={2*{wn}
    }*{l}} ad={2*{wn}*{l}} ps={2*({wn} + 2*{l})} pd={2*({wn} + 2*{l})
    }
.ends
```

Listing 3: NGSpice netlist for Question 1 (q1.cir)

```
.lib $PDK_ROOT/sky130A/libs.tech/ngspice/sky130.lib.spice tt

.param vdd = 1.8

.param vlow = 'vdd*0.2'
.param vhigh = 'vdd*0.8'
.param vmid = 'vdd*0.5'

vdd vdd 0 dc {vdd}
vin vin 0 pulse(0 {vdd} 0 20p 20p 1ns 2ns)

xnot1 vin vout vdd 0 not1
xnot2 vout nc vdd 0 not1

.include not1.cir

.tran 1ps 3ns 0 1ps

.meas tran trise trig v(vout) val={vlow} rise=1 targ v(vout) val={
    vhigh} rise=1
.meas tran tfall trig v(vout) val={vhigh} fall=1 targ v(vout) val={
    vlow} fall=1

.meas tran tplh trig v(vin) val={vmid} fall=1 targ v(vout) val={vmid
    } rise=1
.meas tran tphl trig v(vin) val={vmid} rise=1 targ v(vout) val={vmid
    } fall=1
```

```

.control
run
set gnuplot_terminal=png/quit

gnuplot q1_vout_vin vin vout
+ title 'vout vs vin'
+ xlabel 'time (sec)' ylabel 'voltage (v)'

dc vin 0 1.8 0.01

set nolegend
gnuplot q1_dc_transfer_curve v(vout) vs v(vin)
+ title 'vout vs vin'
+ xlabel 'vin (v)' ylabel 'vout (v)'

let vout = v(vout)
let vin = v(vin)

let slope = deriv(v(vout))

meas dc vm find vin when vout = vin
meas dc vil find vin when slope = -1 cross = 1
meas dc vih find vin when slope = -1 cross = 2

let nml = vil
print nml

let nmh = 1.8 - vih
print nmh

.endc

```

Listing 4: NGSpice netlist for Question 2 (q2.cir)

```

.lib $PDK_ROOT/sky130A/libs.tech/ngspice/sky130.lib.spice tt

.param vdd = 1.8

.param vlow = 'vdd*0.2'
.param vhigh = 'vdd*0.8'
.param vmid = 'vdd*0.5'

vdd vdd 0 dc {vdd}
vin vin 0 pulse(0 {vdd} 0 20p 20p 1ns 2ns)

xnot1 vin vout vdd 0 not2
xnot2 vout nc vdd 0 not1

.include not2.cir
.include not1.cir

```

```

.tran 1ps 3ns 0 1ps

.meas tran trise trig v(vout) val={vlow} rise=1 targ v(vout) val={
    vhigh} rise=1
.meas tran tfall trig v(vout) val={vhigh} fall=1 targ v(vout) val={
    vlow} fall=1

.meas tran tplh trig v(vin) val={vmid} fall=1 targ v(vout) val={vmid
    } rise=1
.meas tran tphl trig v(vin) val={vmid} rise=1 targ v(vout) val={vmid
    } fall=1

.control
run

set gnuplot_terminal=png/quit

gnuplot q2_vout_vin vin vout
+ title 'vout vs vin'
+ xlabel 'time (sec)' ylabel 'voltage (v)'

dc vin 0 1.8 0.01

set nolegend
gnuplot q2_dc_transfer_curve v(vout) vs v(vin)
+ title 'vout vs vin'
+ xlabel 'vin (v)' ylabel 'vout (v)'

let vout = v(vout)
let vin = v(vin)

let slope = deriv(v(vout))

meas dc vm find vin when vout = vin
meas dc vil find vin when slope = -1 cross = 1
meas dc vih find vin when slope = -1 cross = 2

let nml = vil
print nml

let nmh = 1.8 - vih
print nmh

.endc

```

Comments

I initially started with $W_p = 2 \times W_n$ and then adjusted W_p until I obtained approximately equal rise and fall times for the inverter.

For the INVX2 design, I doubled the widths of both the NMOS and PMOS devices compared to the INVX1. After that, I fine-tuned W_p again to achieve balanced rise and fall times.

The results show that the rise time and fall time of the INVX2 are nearly half of those of the INVX1, which matches the expected behavior. The small deviation from the ideal factor of two is due to non-idealities in the circuit and device characteristics.